ModelIt!

Teacher's Guide

Could We Actually Mine Asteroids?

Modeling Space Resource Economics and Orbital Mechanics

Quick Reference

Grade Level	Grade 11L1
NGSS Standard	HS-ESS1-4, HS-ETS1-3
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-ESS1-4, HS-ETS1-3

Use mathematical or computational representations to predict the motion of orbiting objects in the solar system. Evaluate a solution to a complex real-world problem based on prioritized criteria and trade-offs that account for a range of constraints.

Three-Dimensional Learning

Science Practice	**Using Mathematics and Computational Thinking** — Students use the Tsiolkovsky rocket equation and orbital mechanics calculations to predict mission feasibility and evaluate economic trade-offs between delta-v, fuel mass, and resource value.
Disciplinary Core Idea	**ESS1.B Earth and the Solar System / ETS1.B Developing Possible Solutions** — The solar system contains diverse bodies with varying compositions. Engineering design requires evaluating solutions against prioritized criteria and constraints including cost, technology readiness, and physical laws.
Crosscutting Concept	**Scale, Proportion, and Quantity** — Students grapple with the extreme scales involved in space mining — distances in AU, masses in megatonnes, costs in billions, and the exponential relationship between delta-v and fuel mass that makes proportional thinking essential.

CAST Testing Alignment

MULTIPLE CHOICE:

A spacecraft needs 6 km/s of delta-v to reach Asteroid A (C-type, water-rich) and 9 km/s to reach Asteroid B (M-type, platinum-rich). If fuel requirements increase exponentially with delta-v, which statement best describes the cost comparison? A) Asteroid B costs 50% more, B) Asteroid B costs 3-5x more, C) Asteroid B costs about the same, D) The cost depends only on asteroid size

CONSTRUCTED RESPONSE:

Using your model, explain why mining water from near-Earth asteroids for in-space fuel production might be a more viable first business than mining platinum for return to Earth. Include the role of delta-v, the rocket equation, and market economics in your analysis.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

The asteroid belt and near-Earth space contain resources worth quadrillions of dollars — more platinum, nickel, cobalt, and water than humanity could use in millennia. Yet we haven't mined a single gram from space. The barrier isn't knowledge or desire — it's physics. The Tsiolkovsky rocket equation, formulated in 1903, describes an exponential relationship between velocity change and fuel that makes every kilogram sent to space enormously expensive. Understanding asteroid mining requires understanding this equation and the economic constraints it creates.

Asteroid Composition and Classification

Asteroids are classified by spectral type into three main categories. C-type (carbonaceous) asteroids are the most common (~75% of known asteroids) and contain water ice, organic compounds, and clay minerals — water is their most valuable resource because it can be electrolyzed into hydrogen and oxygen rocket propellant. S-type (silicaceous) asteroids (~17%) contain iron, magnesium silicates, and nickel. M-type (metallic) asteroids (~8%) are the jackpot for rare metals — they are fragments of differentiated protoplanetary cores, enriched in iron, nickel, cobalt, and platinum group metals. A single 500-meter M-type asteroid could contain 175 times more platinum than the entire annual Earth production.

The Tsiolkovsky Rocket Equation

The fundamental equation of spaceflight, derived in 1903: delta-v = v_e * ln(m_initial / m_final). This means the velocity change a rocket can achieve depends on its exhaust velocity and the ratio of its initial mass (with fuel) to its final mass (without fuel). The natural logarithm means fuel requirements increase exponentially with delta-v. To achieve 10 km/s with chemical propulsion (v_e ~ 4.4 km/s), you need a mass ratio of about 10:1 — meaning 90% of your spacecraft is fuel. This is why launch costs dominate space economics.

Delta-v Budget for Asteroid Mining

Reaching low Earth orbit from the ground requires about 9.4 km/s of delta-v — the most expensive step in any space mission. From LEO, reaching a near-Earth asteroid requires 4-7 km/s additional delta-v, depending on the asteroid's orbit. Some NEAs require as little as 3.8 km/s from LEO — less than reaching the Moon's surface (5.7 km/s from LEO). This makes certain near-Earth asteroids among the most accessible destinations in the solar system. The return trip can exploit the asteroid's low gravity (no landing/launch penalty) and aerobraking at Earth to reduce delta-v requirements.

The Economics Challenge

Current launch costs range from $2,700/kg (SpaceX Falcon 9) to $10,000/kg (traditional providers). A realistic asteroid mining spacecraft might mass 10,000-50,000 kg, meaning launch alone costs $27-500 million. Add development, operations, and a 2-5 year mission timeline, and total costs reach $1-10 billion. For platinum return: at $30,000/kg, you'd need to return 33-333 tonnes just to break even — possible from a large M-type asteroid but requiring enormous mining and transport infrastructure. For water mining: selling water as fuel in orbit at $10,000-50,000/kg (competing with Earth launch costs) could be profitable with much smaller quantities, making it the likely first commercial product of space mining.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify asteroid type, orbital distance, mission cost, delta-v requirement, resource value, and return on investment as the key components of the space mining economic system.

E **Establish Relationships**

Students determine that asteroid type and orbital distance are external selection variables, while mission cost depends exponentially on delta-v through the rocket equation, resource value depends on asteroid composition, and ROI emerges from the balance of cost and value.

V **Visualize & Model**

Students build a computational model connecting asteroid selection parameters to mission economics through the Tsiolkovsky rocket equation, visualizing how delta-v drives costs exponentially.

E **Evaluate Outcomes**

Students run water mining, platinum mining, and break-even scenarios to discover that near-Earth water mining is more viable than distant platinum mining and that the rocket equation creates a hard ceiling on mission range.

R **Revise & Extend**

Students add market price elasticity, in-space fuel depots, or alternative propulsion types to explore how infrastructure development and technology advances could change the economics.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Asteroid 16 Psyche contains an estimated $10 quintillion in iron and nickel. That's more than the entire global economy — times ten thousand. So why is nobody mining it?"*

Watch For

Show images of asteroids visited by spacecraft (Bennu, Ryugu, Eros). Ask: If these rocks are worth trillions, what's stopping us? Quick-write predictions.

Slide 2: Learning Objectives

3 min

Say: *"Today we're going to discover why one equation — written in 1903 — determines whether space mining is possible. And it's not what you'd expect."*

Watch For

Have students read objectives. Pre-teach 'delta-v' with the analogy: it's like the total budget of velocity changes your spacecraft can make, and every kilometer per second costs exponentially more fuel.

Slide 3: Big Question

3 min

Say: *"Imagine finding a mountain of gold in your backyard — but it's at the bottom of a 100-mile-deep hole. The gold is real. The hole is the physics of getting to space."*

Watch For

Quick-write: Students estimate what it costs to launch 1 kg to space. Reveal: $2,700-$10,000. Calculate what a glass of water would cost in orbit ($5-10 per glass at those rates).

Slide 4: LEVER Framework

2 min

Say: *"We're building a model that answers a billion-dollar question: which asteroids can we actually afford to mine, and what should we mine them for?"*

Watch For

Preview LEVER steps. Emphasize that the rocket equation is the key constraint — everything else flows from it.

Slide 5: Activity 1: Components

8 min

Say: *"We choose the asteroid type and how far away it is. Everything else — cost, fuel, profitability — is determined by physics and economics."*

Watch For

Guide sorting: Asteroid Type and Orbital Distance are external (mission design choices). Mission Cost, Delta-v, Resource Value, and ROI are internal (determined by physics and market conditions).

Slide 6: Activity 2: Connections

8 min

Say: *"Here's where it gets wild. Delta-v doesn't increase cost linearly — it increases it EXPONENTIALLY. Double the distance doesn't mean double the fuel. It means ten times the fuel."*

Watch For

Show the Tsiolkovsky equation visually. Have students plot fuel mass ratio vs. delta-v on graph paper. The exponential curve should shock them.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's test three missions: cheap water mining nearby, expensive platinum mining far away, and a break-even search. Which one makes money?"*

Watch For

Students predict which scenario is profitable. Run simulations. Key insight: water mining nearby beats platinum mining far away because of the exponential fuel cost.

Slide 8: Discoveries

5 min

Say: *"The first asteroid miners won't bring treasure home. They'll sell water as fuel in space. It's not glamorous — but it's the business model that actually works."*

Watch For

Show the exponential cost curve. Compare water-as-fuel economics to platinum-return economics. Discuss the infrastructure-first strategy.

Slide 9: STEM Challenge

12 min

Say: *"You have $2 billion and 7 years. Pick your asteroid, design your mission, prove to investors it makes money. The physics is real. The economics have to be real too."*

Watch For

Groups select target asteroids, calculate delta-v budgets, estimate costs using the rocket equation, and pitch their mission plans. Must address: what to mine, how to make money, and what infrastructure to build.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Asteroid Type, Orbital Distance
- **INTERNAL:** Mission Cost, Delta-v Requirement, Resource Value, Return on Investment

Relationships

- **Orbital Distance to Delta-v Requirement:** POSITIVE (+) — Greater orbital distance requires more velocity changes to reach the asteroid and return, increasing the total delta-v budget. However, the relationship isn't strictly linear because orbital mechanics allows gravity assists and optimal transfer orbits.
- **Delta-v Requirement to Mission Cost:** EXPONENTIAL POSITIVE — The Tsiolkovsky rocket equation creates an exponential relationship: each additional km/s of delta-v requires exponentially more fuel, and that fuel must itself be launched from Earth at $2,700-10,000 per kg. Small increases in delta-v cause massive cost increases.
- **Asteroid Type to Resource Value:** DETERMINES — Asteroid composition directly determines what can be extracted: C-type asteroids provide water (valuable as in-space fuel at $10,000-50,000/kg in orbit), M-type provide platinum group metals (valuable on Earth at $30,000/kg but subject to market crash risk).

Key Discoveries

- The Tsiolkovsky rocket equation means fuel requirements increase exponentially with delta-v — doubling the velocity change far more than doubles the fuel needed, making distant asteroids disproportionately expensive to reach
- Water may be the most valuable resource in space, not platinum — water can be split into hydrogen and oxygen for rocket fuel, eliminating the need to launch fuel from Earth's gravity well at $5,000-10,000 per kilogram
- Some near-Earth asteroids require less delta-v to reach than the surface of the Moon, making them surprisingly accessible targets for early space mining missions
- Bringing massive quantities of rare materials back to Earth would collapse their market price — the economic model only works if demand grows to absorb the new supply, or if resources are used in space rather than returned

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

ModelIt!

Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Asteroids are just floating rocks with nothing valuable"	Many asteroids are remnants of differentiated protoplanetary bodies whose cores concentrated heavy metals through billions of years of geological processes. M-type asteroids — metallic fragments of these ancient cores — contain iron, nickel, cobalt, and platinum group metals in concentrations far exceeding the richest Earth mines. C-type asteroids contain water ice and organic compounds that are invaluable for space operations. The composition of asteroids is well-characterized through spectroscopic observations from Earth and confirmed by spacecraft missions like OSIRIS-REx and Hayabusa2.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Space mining is pure science fiction with no realistic timeline"	Multiple companies (AstroForge, TransAstra, Karman+) are actively developing asteroid mining technology, and NASA's OSIRIS-REx mission successfully returned an asteroid sample in 2023, demonstrating the fundamental capability. SpaceX's Starship — if successful — would reduce launch costs to ~$100/kg, fundamentally changing the economics. The first commercial space mining mission could occur within 10-15 years. The physics works; the engineering is advancing; the remaining challenge is primarily economic.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"You could just send astronauts to mine asteroids like in the movies"	Human asteroid mining is enormously more expensive and dangerous than robotic mining. Sending humans requires life support, radiation shielding, food, water, and return capability — multiplying mission mass and cost by 10-100x. Robotic systems can operate in vacuum, extreme temperatures, and radiation without life support, and can be designed as one-way missions (no return fuel needed for the robot itself). Every serious asteroid mining proposal uses autonomous robots, not astronauts. Humans might supervise remotely, but the mining itself is robotic.